Here are the solutions to the questions from the **University of Barisal Final Examination** for the course **Computer Peripherals and Interfacing (CSE-3207)**, presented in an academic and detailed manner with necessary diagrams.

📝 Final Examination Solution: Peripherals and Interfacing (CSE-3207)

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1. Fundamentals of Peripherals and Interfaces

1. a) What is Interface? Briefly explain different types of Interface. (5 Marks)

An **Interface** is a communication bridge or shared boundary that enables two dissimilar components of a computer system to interact and exchange data. Its primary function is to resolve the mismatch in speed, data format, and control signalling between the high-speed internal components (Processor and RAM) and the typically slower external **Peripherals**[2].

Types of Interfaces:

1. **Intermediate Hardware Interface:** These are physical circuit boards or integrated components that connect directly to the system bus (e.g., PCI or PCIe) and contain the control logic necessary to manage a peripheral.
    - *Examples:* **Nvidia GPU card** [4], **Creative Sound Blaster card** [5], USB Controller [6], Timer, DMA Controller[7].
2. **Intermediate Software/Program Interface:** These are programs, primarily **Device Drivers**, that run on the operating system (OS) kernel and translate high-level I/O requests from applications into the low-level signal sequences and register commands required by the hardware interface[8888].
    - *Examples:* **Nvidia GPU driver** [9], **Sound Blaster Driver software** [10], Linux Device Driver.

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1. b) What do you mean by Computer peripheral? List down the input and output peripherals. (5 Marks)

A **Computer Peripheral** is any device external to the Central Processing Unit (CPU) and main memory (RAM) that expands the computer system's functionality[12]. Peripherals handle input, output, or auxiliary storage functions[13][13].

Category	Definition	Examples
Input Peripherals	Devices used to provide data and control signals to the computer system.	Keyboard, Mouse, Scanner, Webcam.
Output Peripherals	Devices used to display or present information processed by the computer system.	**HD monitor** [14], **5.1 speaker**[15], Printer.
Storage/I/O Peripherals	Devices used for both reading and writing data.	Disk (Hard Drive) [16], USB Drive, Ethernet (Eth)[17].

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1. c) Differentiate between off-line and on-line peripherals. (2 Marks)

Feature	Off-line Peripherals	On-line Peripherals
Connection	Not under the direct control of the CPU or system; data is transferred via an intermediate medium.	Under the **direct and constant** control of the CPU/OS; data is processed in real-time.

Operation	Operates independently or requires human intervention to transfer data to the main system.	Works interactively and dynamically with the processing unit.
Example	Punch card reader (historically), external data logger.	Monitor, Keyboard, Printer, Hard Disk.

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1. d) Write short note on Universal Serial Bus. (2 Marks)

Universal Serial Bus (USB) is an industry-standard interface developed to replace various slower and less versatile interfaces (like parallel and serial ports) with a single, fast, hot-pluggable connection. The USB 3.0 version, also known as the **"SuperSpeed" bus**, offers a transfer mode up to **5.0 Gbit/s** (400 MB/s)[18][18]. It provides data transfer and power delivery to peripherals and uses a tiered-star topology managed by a host controller[19].

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2. CPU Architecture and Digital Circuits

2. a) Define the terms data path and control path. Explain their roles in the execution of instructions and how they interact with each other within a CPU. (4 Marks)

- **Data Path:** The arrangement of hardware elements—such as **Registers**, the **Arithmetic Logic Unit (ALU)**, and internal buses—that handle the flow of data and perform the necessary calculations during instruction execution. The data path is responsible for all data manipulation.
- **Control Path (Control Unit):** The sequential digital logic circuitry that interprets instructions and generates the necessary **timing and control signals** to activate the correct components in the data path and memory system. The control path is responsible

for the overall sequencing and orchestration of operations.

Interaction: The **Control Path** reads the instruction's opcode (via the Instruction Register) and determines the sequence of operations. It then issues precise control signals to the **Data Path** (e.g., enabling registers, selecting the ALU operation, controlling bus multiplexers) to ensure operands flow correctly into the ALU, the operation is executed, and the result is written back to the appropriate destination.

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2. b) Explain the components and functionality of an adder and a multiplexer in digital circuits. (4 Marks)

- **Adder:**
  - **Components:** An adder is constructed from fundamental logic gates (AND, OR, XOR) to perform binary addition. The basic building block is the **Full Adder**, which takes three inputs: two bits to be added (A and B) and a **Carry-in** (C_{in}), producing a **Sum** (S) and a **Carry-out** (C_{out}).
  - **Functionality:** It is the essential component of the **ALU** (Arithmetic Logic Unit) [20] used for addition, subtraction (via two's complement), and address generation. Multi-bit adders are formed by cascading multiple Full Adders (e.g., Ripple Carry Adder).

- **Multiplexer (MUX):**
  - **Components:** A MUX consists of data inputs (2^n), a single output, and n **Select Lines**.
  - **Functionality:** It acts as a **data selector**. The binary value applied to the n select lines determines which one of the 2^n data inputs is routed to the single output. MUXes are vital in CPU design for selecting the appropriate operand source for the ALU or directing data onto a common bus.

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2. c) Define control hazards, data hazards, and structural hazards in the context of pipelined CPU architectures. Discuss the impact of each type of hazard on instruction execution and the techniques used to mitigate them. (4 Marks)

A **hazard** is a condition in an instruction pipeline that causes a stall, preventing the next instruction from executing in the subsequent clock cycle.

Hazard Type	Definition	Impact on Execution	Mitigation Techniques
Control Hazard	Occurs due to **conditional or unconditional jumps** (branches), making it uncertain which instruction to fetch next.	Causes the pipeline to fetch instructions incorrectly down the wrong path, requiring them to be **flushed** and creating large stalls.	**Branch Prediction** (guessing the outcome) and **Delayed Branch** (placing a useful instruction after the branch).
Data Hazard	Occurs when an instruction requires an operand that has not yet been computed and written back by a preceding instruction (Read-After-Write dependency).	Requires the pipeline to **stall** (insert NOP bubbles) until the required data is available.	**Data Forwarding** or **Bypassing** (routing the result directly from the ALU output back to the input of a subsequent instruction).
Structural Hazard	Occurs when two different instructions attempt to use the **same hardware resource** (e.g., memory port, register file port, or ALU unit) simultaneously.	Forces the second instruction to **stall** until the resource is freed by the first instruction.	**Resource Duplication** (e.g., separate instruction and data caches) or **Pipelining** the resource access.

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3. I/O and Control

3. a) What do you mean by the accuracy of an optical mouse? Describe the factors that affect the quality of an optical mouse. (4 Marks)

The **accuracy** of an optical mouse refers to its fidelity in translating physical movement across a surface into precise, corresponding cursor movement on the screen. It is quantitatively measured by its **DPI (Dots Per Inch)** or **CPI (Counts Per Inch)**, which is the number of distinct position changes the sensor can detect per inch of physical movement.

Factors Affecting the Quality/Accuracy of an Optical Mouse:

1. **Resolution (DPI/CPI):** Higher DPI means the mouse can detect smaller movements, leading to greater sensitivity and precision, allowing the cursor to cross the screen with minimal physical movement.
2. **Polling Rate (Hz):** This is the frequency at which the mouse sends its position data to the computer. A higher polling rate (e.g., 1000 Hz) reduces lag and makes tracking smoother, as updates are sent more frequently.
3. **Sensor Technology:** Laser sensors generally offer better accuracy and can work on a wider range of surfaces (including glossy ones) compared to traditional LED-based optical sensors.
4. **Surface Quality:** The texture and uniformity of the mousepad or surface are crucial, as the sensor relies on tracking surface features. Highly reflective, transparent, or overly uniform surfaces can degrade tracking performance.

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3. b) Write short notes on: (4 Marks)

i. **Polling rate:** The **polling rate** is the frequency (measured in Hertz, Hz) at which an input peripheral, such as a mouse or keyboard, reports its current status or position data to the host computer system. A higher polling rate (e.g., 1000 Hz) indicates lower latency, as the system receives position updates more often, leading to a more responsive and precise user experience.

ii. **Refresh rate:** The **refresh rate** (measured in Hertz, Hz) is the number of times per second a display device redraws the image on the screen. For example, a 60 Hz refresh rate means the image is updated 60 times every second. A higher refresh rate reduces motion blur and flicker, contributing to smoother on-screen motion and reduced eye strain.

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3. c) Can a mouse's sensitivity (DPI) be adjusted, and why might someone want to change it? (4 Marks)

Yes, a mouse's sensitivity (DPI) can be adjusted.

Methods of Adjustment:

- **Hardware Buttons:** Many gaming and professional mice feature dedicated DPI buttons that allow users to cycle through preset sensitivity levels instantly.
- **Software/Drivers:** Mouse configuration software allows users to precisely define and store multiple custom DPI settings.
- **Operating System Settings:** OS pointer speed settings apply a multiplier to the mouse's input, effectively changing the on-screen sensitivity.

Reasons for Changing DPI:

- **Gaming:** Gamers often use **High DPI** for rapid cursor movement (e.g., turning quickly in FPS games) and **Low DPI** for precision tasks (e.g., fine aiming/sniping).
- **High-Resolution Displays:** When using high-resolution monitors (e.g., 4K) or multiple-monitor setups, **High DPI** is necessary to move the cursor efficiently across the large screen area without excessive physical hand movement.
- **Precision Work:** Users engaged in graphic design, Computer-Aided Design (CAD), or video editing often use **Low DPI** to gain pixel-level control for accuracy.

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4. Arithmetic and Control in CPU

4. a) Define the function and importance of the Arithmetic Logic Unit (ALU) in a CPU. What types of operations does the ALU perform, and how does it interact with other components of the processor during instruction execution? (4 Marks)

- **Function and Importance:** The **Arithmetic Logic Unit (ALU)** is the combinational digital electronic circuit that performs all the arithmetic and logical operations requested by the program. It is the core execution engine of the processor, as almost every instruction requires the ALU for data manipulation or address calculation[21].

- **Operations Performed:**
  1. **Arithmetic:** Addition, Subtraction, Increment, Decrement.
  2. **Logic:** AND, OR, NOT, XOR.
  3. **Shifting/Rotating:** Bit shifting (logical and arithmetic) and rotation operations.
- **Interaction during Execution:**

1. **Inputs:** The ALU receives its operands from the **General-Purpose Registers (GPRs)** or the **Memory Data Register (MDR)** via the internal data buses.
2. **Control:** The **Control Unit** generates control signals based on the instruction's opcode, telling the ALU *which* specific operation to perform.
3. **Outputs:** The result is typically written back to a GPR or the MDR. The ALU also updates the **Status/Flags Register** (e.g., setting the Zero Flag, Carry Flag) based on the result, which is used by the Control Unit for conditional branching.

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The **Arithmetic Logic Unit (ALU)** is a core functional module within the Central Processing Unit (CPU) responsible for performing all arithmetic and logical operations required during instruction execution. It is widely regarded as the *computational heart* of the processor, as the majority of machine-level instructions eventually rely on the ALU to manipulate binary data. Without the ALU, the CPU would be incapable of executing numerical calculations or making logical decisions, both of which are fundamental to program execution.

Function and Importance of the ALU

The primary function of the ALU is to **execute arithmetic and logic operations** on the data supplied to it by the processor's registers or memory. It plays a crucial role in transforming raw binary data into meaningful computational results. Its importance can be summarized as follows:

1. **Essential for Data Processing:**
   All mathematical calculations, logical comparisons, and bit-level manipulations required by programs are executed through the ALU. This enables the CPU to perform computations required in applications such as scientific computing, encryption, graphics, and general-purpose processing.

2. **Supports Decision-Making:**
   The ALU generates **status flags** (e.g., Zero, Carry, Overflow, Sign), which are stored in the processor's status register. These flags guide the CPU in making decisions during conditional branching and program control flow.

3. **Integral to the Instruction Execution Cycle:**
   During the *execute* phase of the instruction cycle, the ALU processes the operands and produces the final result, which is then stored back in a register or memory. Thus, it is indispensable for completing instruction execution.

Types of Operations Performed by the ALU

The ALU supports several categories of operations, including:

1. **Arithmetic Operations:**
    - Addition, subtraction, multiplication, and division
    - Increment and decrement
    - Modulus and comparison of numerical values
2. **Logical and Bitwise Operations:**
    - AND, OR, NOT, NAND, NOR, XOR, XNOR
    - Bit shifting (logical and arithmetic shifts)
    - Bit rotation (left rotate, right rotate)
3. **Relational/Comparison Operations:**
    - Equal-to, less-than, greater-than, not-equal
      These operations contribute to decision-making within programs.

Interaction of the ALU with Other CPU Components

The ALU does not operate independently; it collaborates closely with other processor components to execute instructions efficiently:

1. **Interaction with Registers:**
   The operands required for operations are retrieved from **general-purpose registers** (e.g., R1, R2). After computation, the result is written back to a register or forwarded to memory.

2. **Interaction with the Control Unit (CU):**
   The CU sends **control signals** specifying which ALU operation to perform. It orchestrates the timing and coordination of ALU operations within the instruction cycle.

3. **Interaction with Internal Buses:**

The ALU receives input data and sends output results via the CPU's internal data buses, ensuring seamless data transfer between memory, registers, and the ALU.

4. **Interaction with the Status/Flag Register:**
   After each operation, the ALU updates condition flags (Zero, Carry, Overflow, Negative) which influence the behavior of subsequent instructions such as conditional jumps and loops.

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4. b) Create a flowchart to illustrate the process of multiplying two numbers using assembly language instructions. Explain the key steps involved in the flowchart and how they translate into assembly language instructions. (4 Marks)

The simplest assembly method for multiplication is through **repeated addition**.

Flowchart: Multiplication ($A \times B$) via Repeated Addition

1. **Start**
2. **Initialize:** Set Result $\text{R}=0$. Set Counter $\text{C}=0$.
3. **Check Condition:** Is $\text{C} < \text{B}$? (Is the multiplier B exhausted?)
4. **If YES (Loop):** $\text{R} \leftarrow \text{R} + \text{A}$ (Add Multiplicand).
5. **Increment:** $\text{C} \leftarrow \text{C} + 1$.
6. **Loop:** Go back to **Check Condition** (Step 3).
7. If NO (Exit): End (Result is in R).
   *
   Key Steps and Assembly Translation (Conceptual):

Step	Explanation	Conceptual Assembly Code
Initialization	Set result register (R0) and counter register (R1) to zero.	MOV R0, #0 (Result) MOV R1, #0 (Counter)
Condition Check	Compare counter to multiplier (R2) and branch if equal/greater.	LOOP: CMP R1, R2 (Compare R1 and R2) BGE END (Branch if Greater or Equal to END)

Addition	Add the multiplicand (R3) to the result register (R0).	ADD R0, R0, R3
Increment/Loop	Increment the counter (R1) and unconditionally jump back to the start of the loop.	INC R1 B LOOP

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4. c) Define the terms Stack Pointer (SP) and Stack Memory in the context of computer architecture. (4 Marks)

- **Stack Memory:**
  - **Definition:** A specific segment of Random Access Memory (RAM) reserved for temporary storage that operates on a **Last-In, First-Out (LIFO)** principle.
  - **Function:** It is crucial for managing subroutine or function calls by storing the **return address**, holding the values of local variables, and saving the contents of general-purpose registers during context switches (e.g., interrupts).
- **Stack Pointer (SP):**
  - **Definition:** A dedicated CPU register that contains the **memory address** of the current top element of the stack.
  - **Function:** The SP controls stack operations. The PUSH operation typically **decrements** the SP (as the stack usually grows downwards in memory) and then writes data to the new address. The POP operation reads the data from the address pointed to by SP, and then **increments** the SP.

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5. Performance Metrics and Registers

5. a) Define cycle time and CPI (Cycles Per Instruction). How do these metrics influence processor performance? Provide an example of how reducing CPI or cycle time can improve overall performance. (6 Marks)

- **Cycle Time (τ):**
  - **Definition:** The time duration of one clock cycle, which is the reciprocal of the processor's clock frequency (f). $\tau = \frac{1}{f}$.

    - **Influence:** Reducing τ (i.e., increasing clock frequency) directly decreases the total time required for all operations, leading to faster instruction execution.
- **CPI (Cycles Per Instruction):**
    - **Definition:** The average number of clock cycles required by the CPU to execute a single instruction in a given program.
    - **Influence:** A lower CPI indicates a more efficient architecture or pipeline, meaning fewer wasted cycles per instruction, which reduces total execution time.
- **Overall Performance (Execution Time):** Both metrics are related by the fundamental CPU performance equation:
  $$ \text{Execution Time} = (\text{Instruction Count}) \times \text{CPI} \times \text{Cycle Time}$$
  Reducing any of the terms (Instruction Count, CPI, or Cycle Time) will lead to an improvement in overall performance (lower execution time).
- Example:
  Consider a program with 100 instructions.
    - **Scenario A:** $\text{CPI}=4$, $\tau=2 \text{ ns}$. $\text{Time} = 100 \times 4 \times 2 \text{ ns} = 800 \text{ ns}$.
    - Scenario B (Reduced CPI): $\text{CPI}=2$, $\tau=2 \text{ ns}$. $\text{Time} = 100 \times 2 \times 2 \text{ ns} = 400 \text{ ns}$.
      Reducing the CPI from 4 to 2 (e.g., through improved pipeline efficiency) halves the execution time.

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5. b) Define registers in the context of computer architecture. What are the different types of registers, and how do they contribute to the overall performance of a CPU? (6 Marks)

- **Definition: Registers** are small, very high-speed temporary storage locations residing within the CPU itself. They hold data, addresses, or control information that is currently being used by the processor. Registers are the fastest form of memory available to the CPU.
- **Types of Registers and Performance Contribution:**

Register Type	Definition/Function	Contribution to Performance
General Purpose Registers (GPRs)	Store operands, variables, and intermediate results for the ALU[22].	**Speed/Efficiency:** By keeping frequently accessed data in GPRs, the CPU avoids the much slower latency of accessing

		main memory (RAM).
Program Counter (PC)	Stores the memory address of the **next** instruction to be fetched and executed.	**Flow Control:** Essential for sequential instruction execution and for directing the processor to the correct address for branches/jumps.
Instruction Register (IR)	Holds the instruction currently being executed or decoded by the Control Unit.	**Pipelining:** Critical for the instruction decode stage, allowing the PC to fetch the next instruction simultaneously, enabling pipeline parallelism.
Memory Address/Data Registers (MAR/MDR)	MAR holds the address for memory access; MDR holds the data being transferred to/from memory.	**Interfacing:** Buffers high-speed CPU operations from the slower memory bus, optimizing memory transaction speed.

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6. Instruction Set Architecture

6. a) Define processor architecture and explain its importance in computer systems. Describe the key components of a processor architecture and how they work together to execute instructions. (6 Marks)

- **Processor Architecture (Instruction Set Architecture - ISA):**
  - **Definition:** The abstract model of a computing system that specifies all the programmer-visible elements, including the set of instructions (opcodes), the addressing modes, the processor registers, the native data types, and how the memory is addressed.
  - **Importance:** It is the **fundamental hardware/software interface**. It dictates how software (OS, compilers, drivers) must be written (i.e., the machine language) to run

correctly on the processor's hardware implementation.

- **Key Components and Instruction Execution:**
  1. **Registers:** Store the current state (PC, SP, GPRs).
  2. **Control Unit: Fetches** the instruction from memory (address from PC), **decodes** it (in the IR), and generates control signals.
  3. **ALU:** Performs the required **arithmetic or logical operation** on the operands provided by the registers.
  4. Internal Buses: Provide the data paths for moving instructions from the IR and operands between Registers and the ALU.
     The Control Unit orchestrates these components through the Fetch-Decode-Execute-Writeback cycle to complete an instruction.

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6. b) Compare RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing) architectures. What are the advantages and disadvantages of each, and in what types of applications are they typically used? (6 Marks)

Feature	**RISC (Reduced Instruction Set Computing)**	**CISC (Complex Instruction Set Computing)**
Instruction Set	**Small** and uniform instruction format; simple operations (mostly register-to-register).	**Large** and variable instruction format; complex operations (e.g., single instruction loads data from memory, performs an operation, and stores the result).
Clock Cycles	Focuses on minimizing CPI (Cycles Per Instruction), ideally 1 cycle per instruction.	Higher CPI due to complex, multi-cycle instructions.
Registers	Large number of registers.	Smaller number of registers.
Advantages	Simple design facilitates	High **Code Density**

	pipelining; easier to achieve high clock frequencies; lower power consumption.	(complex tasks use fewer instructions); simpler compiler development (historically).
Disadvantages	Compiler must generate more instructions for complex tasks (lower code density).	Difficult to pipeline efficiently; complex control unit design.
Typical Use	**Mobile Devices (ARM), Embedded Systems, Routers, Servers:** Where power efficiency and high throughput are critical.	**Desktops/Laptops (x86/Intel/AMD):** Legacy compatibility and historical dominance.

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7. Peripherals and Modern Interfaces

7. a) Explain the architecture and performance of GPU. Why are GPUs particularly well-suited for parallel processing tasks? (4 Marks)

- **GPU Architecture and Performance:**
  - A **GPU (Graphics Processing Unit)** features a highly **parallel** architecture comprising hundreds or thousands of small, specialized processing cores (like the **480 Cuda cores** in the Nvidia GTX295)[23232323].
  - Its design prioritizes **throughput** (completing a large volume of data operations) over **latency** (the speed of a single operation). The cores are grouped into streaming multiprocessors that share control logic.
- Suitability for Parallel Processing:
  GPUs are uniquely efficient for parallel tasks (e.g., graphics rendering, scientific simulations, AI/ML) because:
  1. **SIMD (Single Instruction, Multiple Data):** They are optimized for executing the **same instruction** across a **large dataset** simultaneously, which is typical of graphics and matrix operations.
  2. **High Thread Count:** The vast number of cores allows the GPU to launch thousands

of threads concurrently, effectively hiding the memory access latency by switching execution to a ready thread whenever another thread stalls.

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7. b) Discuss the advancements introduced with Bluetooth 4.0 technology in comparison to its predecessors. What key features made Bluetooth 4.0 a significant improvement for mobile devices? (4 Marks)

Bluetooth 4.0 (launched around 2010) was a pivotal advancement because it introduced the **Bluetooth Low Energy (BLE)** specification alongside Classic Bluetooth[242424242424242424].

Feature	Improvement in Bluetooth 4.0 (BLE)
Power Consumption	**Ultra-Low Energy:** Designed for devices that send small packets of data infrequently, allowing them to run for months or years on a coin cell battery.
Power Management	Faster connection establishment and lower duty cycle (sleep most of the time).
Key Features for Mobile Devices:	
Wearables/IoT:	It enabled the proliferation of **wearable technology** (fitness trackers, smartwatches, sensors) by providing a reliable, low-power connection that didn't drain the host mobile device's battery[25].
Dual Mode:	Mobile devices could implement a **dual-mode** chip, supporting both the high-bandwidth needs of **Classic Bluetooth** (for audio streaming) and the power-saving feature of **BLE**.

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7. c) USB 3.0, launched in 2010, introduced key improvements over earlier versions. Explain its main features and benefits. (4 Marks)

USB 3.0, branded as **"SuperSpeed"**, introduced major architectural and performance improvements over USB 2.0[26].

Main Features and Benefits:

1. **SuperSpeed Transfer Rate:** The primary feature is its maximum theoretical transfer rate of **5.0 Gbit/s**, which is equivalent to approximately **400 MB/s**[27]. This is about 10 times faster than USB 2.0, significantly speeding up transfers to devices like external SATA HDDs[28]. The high speed is enabled by technologies like **8B/10B encoding** and **scrambling**[29].

2. **Full-Duplex Communication:** USB 3.0 added a new physical bus structure dedicated to SuperSpeed, allowing data to be transmitted and received **simultaneously** (full-duplex), which boosts overall I/O efficiency.
3. **Increased Power and Better Management:** It increased the available power for bus-powered devices and introduced better power management states, allowing devices to enter low-power modes when idle and wake up faster.

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8. Ergonomics and Human Interface

8. a) Write short notes on the Hall effect keyboard. (4 Marks)

A **Hall Effect Keyboard** uses the **Hall effect** principle—the generation of a voltage difference across an energized conductor when a magnetic field is applied perpendicular to the current flow—to detect keystrokes.

- **Mechanism:** Each keycap has a magnet attached underneath it. As the key is pressed, the magnet moves closer to a stationary Hall effect sensor embedded in the PCB. The change in the magnetic field magnitude is sensed by the sensor, which triggers a

keypress signal when a set threshold is crossed.

- **Advantages:** They are highly **durable and long-lasting** because there are **no physical contacts** to wear out (unlike mechanical switches). This non-contact mechanism also allows for an adjustable actuation point in some models, meaning the user can customize how far the key must travel before it registers a press.

Working Principle of Hall Effect Keyboards

1. **Magnet Movement with Key Press**
    - Each key has a **small magnet** attached underneath.
    - When the key is pressed, the magnet moves closer to a **Hall effect sensor** located on the circuit board.
2. **Hall Effect Sensing**
    - The Hall effect sensor detects the **magnetic field** produced by the magnet.
    - According to the **Hall effect principle**, when a magnetic field is applied **perpendicular** to a current flowing through a conductor, a **voltage is generated at a right angle** to both the current and the magnetic field.
    - This voltage change is the basis for detecting a key press.
3. **Signal Generation**
    - As the magnet's proximity changes, the sensor produces a **voltage signal**.
    - The keyboard's controller processes this signal to identify **which key was pressed**.
4. **Durability Advantage**
    - Because the detection relies on **magnetic fields rather than physical contact**, there is **minimal wear and tear**.
    - This design allows Hall effect keyboards to withstand **millions of key presses** without degradation, making them highly durable compared to dome switch or mechanical contact-based keyboards.

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8. b) Write down the difference between laser keyboard and dome switch keyboard. (4 Marks)

Feature	Laser Keyboard (Virtual Projection)	Dome Switch Keyboard (Rubber Dome)
Mechanism	Projects a keyboard image onto a flat surface using a laser and uses optics to track finger position to register a keystroke.	Uses a silicone or rubber dome under each key. Pressing the key collapses the dome, pushing a conductive pad against a circuit membrane.
Form Factor	Extremely compact and portable; no physical keys or travel distance.	Traditional physical keys with tactile travel and defined spacing.
Typing Feel	No physical feedback (typing on a hard surface); accuracy is generally low for touch typists.	Soft, quiet, and "mushy" tactile feedback from the collapsing dome.
Use Case	Novelty or very light use with mobile devices where portability is paramount.	Most common, cost-effective keyboard found in offices and homes.

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8. c) Discuss the impact of proper keyboard ergonomics on user productivity and health. What are some key ergonomic tips for reducing strain when using a keyboard for long periods? (4 Marks)

- **Impact on User Productivity and Health:**
  - **Health:** Poor keyboard posture (e.g., extended or bent wrists, hunched shoulders) leads to **Musculoskeletal Disorders (MSDs)**, most notably **Repetitive Strain Injuries (RSIs)** like Carpal Tunnel Syndrome and Tendinitis. These conditions cause chronic pain and long-term injury.
  - **Productivity:** Discomfort and pain necessitate frequent breaks, slow down typing speed, and increase the likelihood of errors, severely reducing sustained productivity.
- **Key Ergonomic Tips for Reducing Strain:**
  1. **Neutral Wrist Posture:** Keep the wrists straight and flat, avoiding any upward bend (extension) or side-to-side tilt (ulnar deviation).
  2. **Proper Height:** Position the keyboard at a height where the elbows are bent at a 90-degree angle and the shoulders are relaxed.

3. **Floating Wrists:** Avoid resting wrists on the desk or a wrist rest while actively typing; allow the wrists to float above the keyboard.
4. **Ergonomic Keyboard:** Use a **split** or **contoured** ergonomic keyboard to allow the hands to maintain a more natural, shoulder-width position.
5. **Take Breaks:** Implement micro-breaks to stretch hands and arms, reducing cumulative strain.